

SOP-PSM-2500: Process Hazard Analysis

Document owner: Process Safety Department **Effective date:** 2024-06-01 **Review cycle:** 60 months

Section 1. Purpose and Scope

Governs conduct and revalidation of Process Hazard Analysis (PHA) for covered processes under the site's Process Safety Management program.

Section 2. Revalidation Interval

Clause 2.1 — Each covered process's PHA shall be revalidated at least every 5 years using a recognized methodology (HAZOP, What-If, or equivalent).

Clause 2.2 — A significant process change under SOP-MOC-2400 that was not addressed by the current PHA triggers an interim PHA update for the affected node(s) rather than waiting for the next full revalidation.

Section 3. Team Composition

Clause 3.1 — The PHA team shall include an operator or supervisor with process-specific experience, a process engineer, and a facilitator trained in the chosen methodology; a PHA conducted entirely by engineering staff without operations input is not acceptable.

Section 4. Recommendation Tracking

Clause 4.1 — Every PHA recommendation is assigned an owner and resolution date; recommendations shall not be closed as “no action needed” without documented engineering justification reviewed by the Process Safety Department.

Clause 4.3 — A recommendation addressing a safeguard gap for a scenario with potential for serious injury or major loss shall be resolved within 12 months of PHA completion, or an interim safeguard implemented immediately if resolution will take longer.

Section 5. Records

Clause 5.1 — PHA reports, revalidations, and recommendation tracking are filed to the Process Safety File and retained for the life of the process plus the required regulatory retention period.